Mutation of Local Floer Chain Complexes across Hamiltonian Bifurcations

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Contents

Based on a joint work with Hong-kwon Jo (SNU), arXiv:2607.22178.

Main result. We describe the explicit changes of the Floer chain complex for generic types of Hamiltonian bifurcations.

- **Motivation and Preliminaries.**
  Bifurcations, the Conley–Zehnder index, and Floer homology.
- **Methodology.**
  Model Hamiltonian, Morse–Bott type cascade counting.
- **Chain complex mutation.**
  Result for 6 types of bifurcations.
  (Generic 4 types, $\mathbb{Z}_2$-symmetric 2 types.)

Motivation

Motivation 1. Bifurcations

A **bifurcation** is, heuristically, the birth or death of periodic orbits.

Several recent works [AFvK$^+$23], [AB25], [JKvK26] relate bifurcations to symplectic invariants, for example Conley–Zehnder indices.

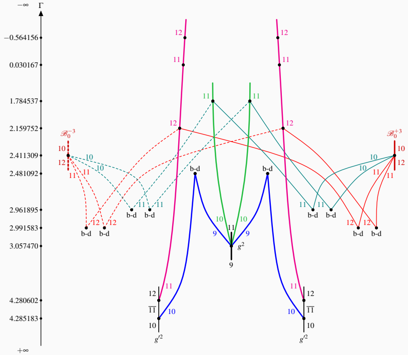


Figure 1: Bifurcation graph of Hill's lunar problem, [AB25].

Can we make a *symplectic dictionary* for bifurcation types?

Motivation 2. Rotating Kepler Problem

In the rotating Kepler problem, there are three important periodic orbits: retrograde, direct and vertical collision orbits.

The multiple covers of the retrograde and direct orbits bifurcate, producing orbits that look like flowers.

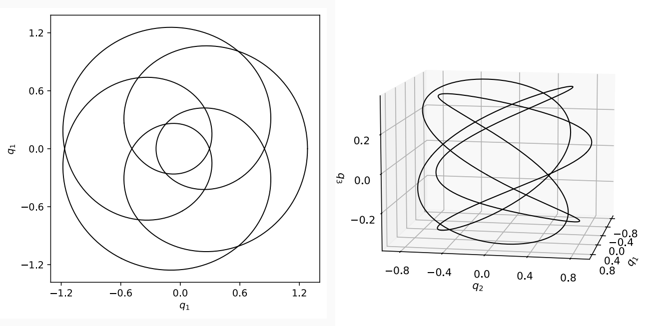
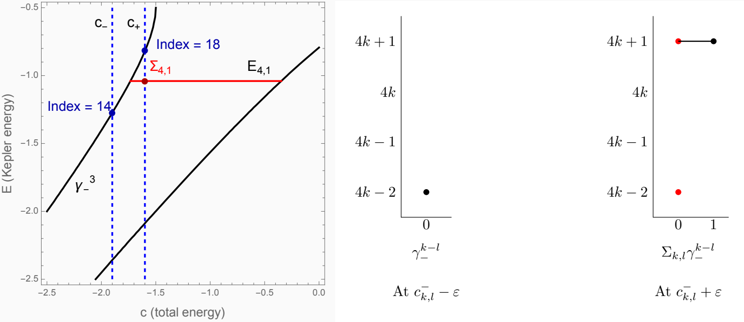


Figure 2: Resonant orbits of RKP, courtesy of Chankyu Joung.

Motivation 2. Rotating Kepler Problem

In my previous work [Lee26], I observed that this bifurcation can be translated into a change in the Floer chain complex.



Slogan. A bifurcation gives *a mutation of a Floer chain complex*.

We aim to describe this effect explicitly.

Periodic Orbits and Bifurcations

Periodic Orbits

Let (W, ω) be a symplectic manifold and $H : W \to \mathbb{R}$ be a function.

We define a **Hamiltonian vector field** by a relation $i_{X_H}\omega = -dH$.

Note. If $W = T^*\mathbb{R}^n$, $\omega = dp \wedge dq$ and $H = |p|^2/2 + V(q)$, X_H gives usual Hamiltonian equation in the classical mechanics,

$$\dot{q} = p, \quad \dot{p} = -V'(q).$$

Main interest is **periodic orbits**, which is an orbit γ such that $\dot{\gamma} = X_H$ and $\gamma(0) = \gamma(T)$ for some $T > 0$.

Periodic orbits form backbone of the dynamics, and also connects the topology and dynamics. (Arnold conjecture, Floer theory)

Poincaré Section

Let γ be a periodic orbit of a Hamiltonian system and $p \in \gamma$.

A **Poincaré section** is a transversal section $\Sigma \subset H^{-1}(c)$ at p.

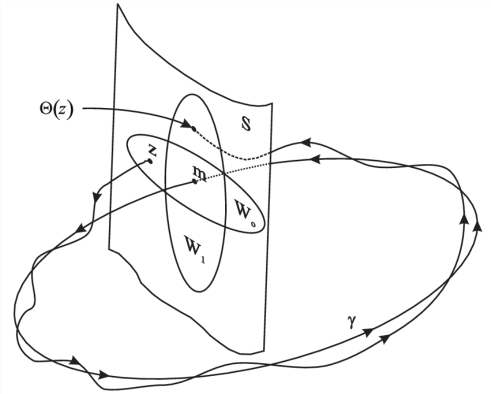


Figure 3: Poincaré section [OR99].

Stability

A **return map** is defined by $\Phi : U \subset \Sigma \to \Sigma$ by

$$\Phi(q) = Fl^{X_H}_{\tau(q)}(q), \quad \tau(q) = \min\left\{t \,:\, Fl^{X_H}_t(q) \in \Sigma\right\}.$$

Floquet multipliers are eigenvalues of $d_p\Phi$.

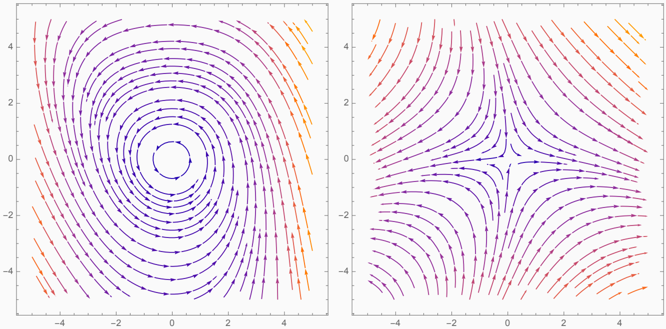
Since $d_p\Phi$ is symplectic, these form a quadruple $\{\lambda, \bar{\lambda}, \lambda^{-1}, \bar{\lambda}^{-1}\}$.

In 4-dimensional autonomous case, we have 3 possibilities:

- $\lambda = \lambda^{-1} \,:\, \lambda \in S^1 \subset \mathbb{C}$, called **elliptic**.
- $\lambda = \bar{\lambda} \,:\, \lambda \in \mathbb{R} \setminus \{0\}$, called **hyperbolic**.
- $\lambda = \pm 1$.

We call this **stability** of a periodic orbit.

Stability



(a) Elliptic **(b)** Hyperbolic

Figure 4: Elliptic and hyperbolic fixed points

Bifurcation

Let $H_\varepsilon : W^4 \to \mathbb{R}$ be a 1-parameter family of Hamiltonians, and let γ_0 be a periodic orbit of H_0 with Floquet multiplier λ_0.

If $\lambda_0 \neq 1$, so the orbit is **non-degenerate**, then nearby periodic orbits form an *orbit cylinder*

$$\Gamma = \{(\gamma_\varepsilon, \varepsilon) : \gamma_\varepsilon \text{ is a periodic orbit of } H_\varepsilon\}.$$

This is a *continuation* of the orbit along the bifurcation parameter.

We follow γ_ε and detect λ_ε. If $\lambda_\varepsilon = 1$ at some ε, then γ_ε is degenerate, and the orbit cylinder may break:

- periodic orbits may be born or vanish; this is called a **bifurcation**,
- if $\lambda_\varepsilon \neq 1$ but $\lambda_\varepsilon^k = 1$, then the k-th cover bifurcates.

Example. Period Doubling

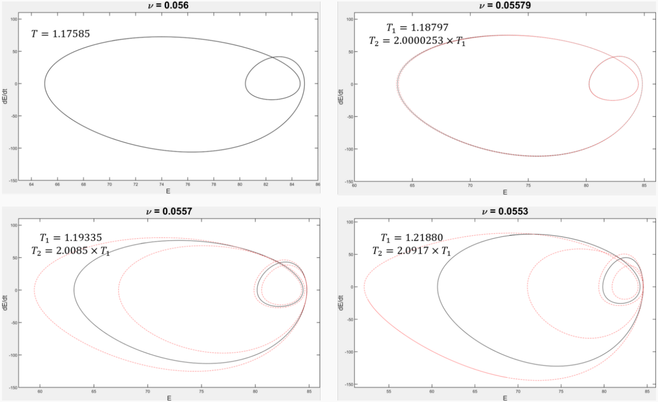


Figure 5: Period doubling in Kuramoto–Sivashinsky equation [Max21].

Period doubling is a generic bifurcation of a double cover.

The red dashed orbit is the new orbit that emerges from the bifurcation.

Conley–Zehnder Index

The **Conley–Zehnder index** $\mu_{\mathrm{CZ}}(\gamma)$ is an integer associated with a periodic orbit and can be understood as a *symplectic winding number*.

We are interested in a change of index along the parameter; $\mu_{\mathrm{CZ}}(\gamma_\varepsilon)$.

Along a non-degenerate family, the CZ index is constant.

Key idea. A change in μ_{CZ} = Degeneracy of γ_ε = A bifurcation occurs

Hence, a bifurcation can be detected via change of the CZ index.

Note. This is also useful in practical purposes. ([AB25], [JKvK26].)

Floer Homology

Floer homology $HF(W,H)$ is a topological invariant associated with a symplectic manifold W and Hamiltonian H. (cf. [AD14].)

This is determined by the **Floer chain complex** $CF(W,H)$, whose

- generators are 1-periodic orbits of H,
- grading is given by CZ indices,
- the differential is determined by counting **Floer cylinder**, whose boundaries are asymptotic to periodic orbits.

Across a bifurcation, $CF(W,H)$ changes substantially;

- orbits are born or vanish $\Rightarrow$ the number of generators changes.
- the CZ index changes $\Rightarrow$ grading changes

However, the homology doesn't change, so the differential is responsible for keeping the homology invariant. *Can we describe this explicitly?*

Schematic Picture

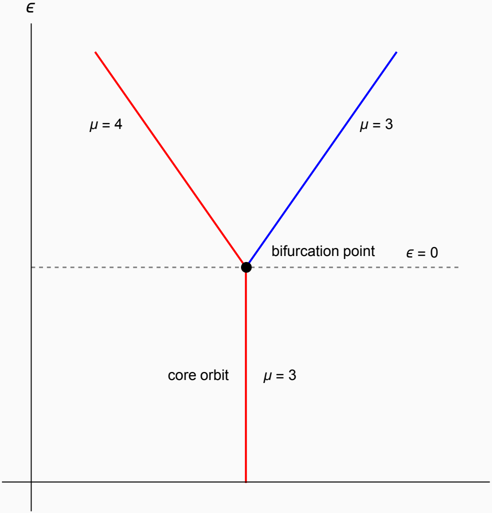


Figure 6: A bifurcation diagram for the period doubling.

Methodology

Main Result and Key Ideas

Main result. A full description of the mutation of Floer chain complex across the seven bifurcation types, and the local Floer homology of degenerate orbits.

Key Ideas.

1. Construct a Hamiltonian H_ε represents a given bifurcation type.
2. Gradient trajectories of G_ε give Floer cylinders of H_ε.
3. Morse indices of critical points of G_ε give relative CZ indices.
4. We get cylinders connecting a simple orbit and a multiple cover.
   $\Rightarrow$ A contribution of 1 cylinder may not be exactly 1.
5. Using Morse–Bott cascades, we can compute the explicit chain complex and verify that its homology is invariant.

Meyer's Classification

Generic Hamiltonian bifurcations are classified in [Mey70].

They are organized by the covering number k of the underlying orbit.

- $k = 1$: birth-death
- $k = 2$: period doubling, materialization (opposite stability types)
- $k = 3, 4$: phantom kiss
- $k \geq 4$: emission

Aside. We will use names from [AM78], where three extra types involving critical points are given.

Other names: touch-and-go, period tripling/quadrupling, etc.

Note. The fourth cover is a borderline case, depending on the coefficients of the normal form.

Birkhoff Normal Form

We find the fixed points of the iterated return map Φ_ε^k.

This can also be done in the essentially same way by finding critical points of the generating Hamiltonian G_ε of Φ_ε^k.

1. If $\lambda = \exp(i\alpha)$, we can find a formal power series $G = \sum G_n$ of the generating Hamiltonian of Φ^k of the form

$$G_t = G_2 + \sum_{n\geq 3} G_{n,t}$$

   where $\deg G_n = n$, $G_2^{ss} = \alpha(u^2+v^2)$, $\dot{G}_{n,t} = \{G_2^{ss}, G_{n,t}\}$.
2. If $\lambda^k = 1$, $G_2^{ss} = (2\pi l/k)(u^2+v^2)$, which means that G is generated by polynomials commuting with rotation by $2\pi l/k$, i.e.,

$$G \in \mathbb{R}[\rho^2, \rho^k \cos(k\varphi - 2\pi l t), \rho^k \sin(k\varphi - 2\pi l t)].$$

3. Using rotating frame, we remove the time-dependency and G becomes an **autonomous $\mathbb{Z}_k$-symmetric polynomials**.

Birkhoff Normal Form

4. Ring of $\mathbb{Z}_k$-symmetric polynomials is $\mathbb{R}[\rho^2, \rho^k \cos k\varphi, \rho^k \sin k\varphi]$.
5. At $\varepsilon = 0$, the semisimple quadratic part should be zero, which means that $G_2 = \alpha x^2$ (shear, $k = 1, 2$) or $G_2 = 0$ $(k \geq 3)$.
6. At $\varepsilon \neq 0$, the leading term of perturbative part has degree 1 $(k = 1)$ or 2 $(k \geq 2)$.

$\Rightarrow$ We can write the Hamiltonians by

$$G = \begin{cases} \alpha x^2 + \beta y^3 + \varepsilon(ax + by + \dots) & k = 1, \\ \alpha x^2 + \beta y^4 + \varepsilon(ay^2 + \dots) & k = 2, \\ \alpha \rho^4 + \beta \rho^k \cos k\varphi + \varepsilon(ar^2 + \dots) & k \geq 3. \end{cases}$$

The critical points of G are fixed points of Φ^k, which is k-periodic points of Φ, so there is a k-**to-1 correspondence** except for the origin.

Model Hamiltonian

Let $\tilde{H}_\varepsilon$ be a Hamiltonian and G_ε be a generating Hamiltonian of Φ_ε^k.

We construct a model Hamiltonian on $I_r \times (\mathbb{R}/k\mathbb{Z})_\theta \times \Sigma_z$ by,

$$H_\varepsilon = kh(r) + G_\varepsilon(z).$$

where $h'(0) = 1$ and $h''(0) \neq 0$.

- H_ε is defined on the covering space of a neighborhood of a periodic orbit, but Floer-type information descends to the base space.
- The rotating number, or CZ index, is encoded in the covering map.
- H_ε is not exactly the same as $\tilde{H}_\varepsilon$, but has the same return map up to finite degree and relative Conley–Zehnder indices as $\tilde{H}_\varepsilon$.
- We also have $\mu_{CZ}(\gamma_p) = 1 - \mathrm{Ind}_{\mathrm{Morse}}(p)$.
- Gradient flow between critical points of G_ε gives a Floer cylinder, which we call a **gradient revolution**.

Morse–Bott Cascade

Our periodic orbits have an S^1-symmetry under time shifts, which makes the situation slightly different from the usual Floer theory.

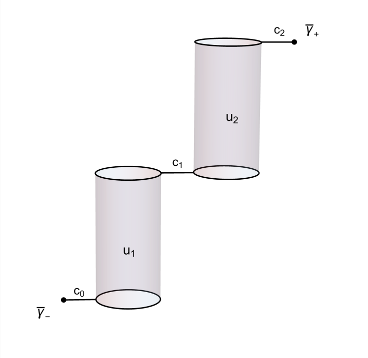
We equip each orbit with a Morse function, and consider two critical points (max $\hat{\gamma}$ and min $\check{\gamma}$) as generators of a Floer chain complex.

In this case, the differential is given by counting **Morse–Bott cascades**; a sequence of Morse trajectories and Floer cylinders.

How to count?

1. $(\mu(\gamma_+;\gamma_-) = 1)$ Morse trajectory on γ_- from $\hat{\gamma}_+$ to $ev^{-1}(\hat{\gamma}_-)$ or from $ev(\check{\gamma}_+)$ to $\check{\gamma}_-$
2. $(\mu(\gamma_+;\gamma_-) = 2)$ Morse trajectory on γ_0 (such that $\mu(\gamma_0;\gamma_-) = 1)$ from $ev(\check{\gamma}_-)$ to $ev^{-1}(\hat{\gamma}_+)$.

Morse–Bott Cascade



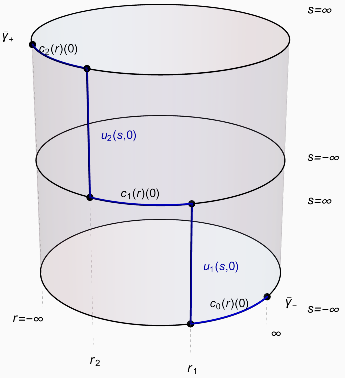


Figure 7: Morse–Bott cascades.

Cascade Counting Theorem

Theorem

Let $\gamma_\pm$ be two simple non-constant $1/k_\pm$-periodic orbits, so that $\gamma_\pm^{k_\pm}$ are 1-periodic orbits. Moreover, assume that

$$\mu_{CZ}(\gamma_+^{k_+}) - \mu_{CZ}(\gamma_-^{k_-}) = 1.$$

Then, the quotient $\mathcal{M}_0(\gamma_-^{k_-}, \gamma_+^{k_+})/S^1$ is a 0-dimensional $\mathbb{Z}_{(k_+,k_-)}$-orbifold, and we have

$$\#_2\mathcal{M}(\hat{\gamma}_-^{k_-}, \hat{\gamma}_+^{k_+}) \equiv k_- \cdot \#(\mathcal{M}_0(\gamma_-^{k_-}, \gamma_+^{k_+})/S^1) \pmod 2,$$
$$\#_2\mathcal{M}(\check{\gamma}_-^{k_-}, \check{\gamma}_+^{k_+}) \equiv k_+ \cdot \#(\mathcal{M}_0(\gamma_-^{k_-}, \gamma_+^{k_+})/S^1) \pmod 2.$$

In particular, if there is only one cylinder from a simple γ_0 to a multiple cover γ_1^k,

$$\partial\hat{\gamma}_0 = k\hat{\gamma}_1^k, \quad \partial\check{\gamma}_0 = \check{\gamma}_1^k.$$

Idea of the proof

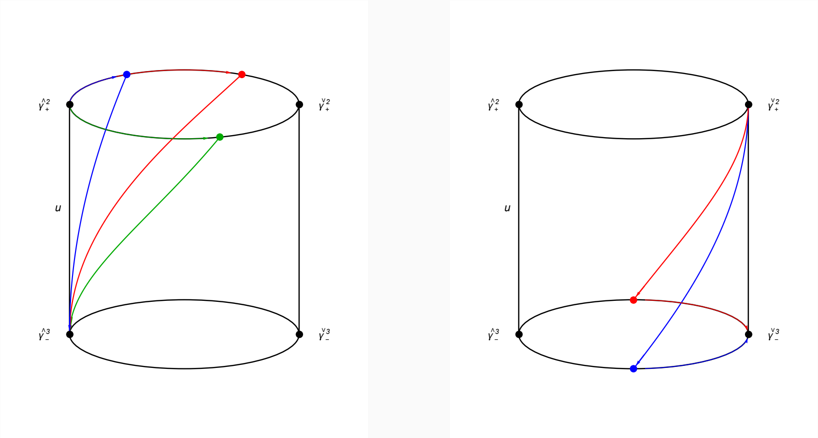


Figure 8: Cascades between 2nd and 3rd covers.

Chain Complex Mutation

Birth-death $(k = 1)$

A generic single cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(u, v) = \alpha v^2 + \beta u^3 + \ldots + \varepsilon(u + \ldots).$$

- $(\varepsilon < 0)$ There are no periodic orbits.
- $(\varepsilon > 0)$ One elliptic γ_0 and one hyperbolic orbit γ_1 are born.
- There is one gradient trajectory between them.

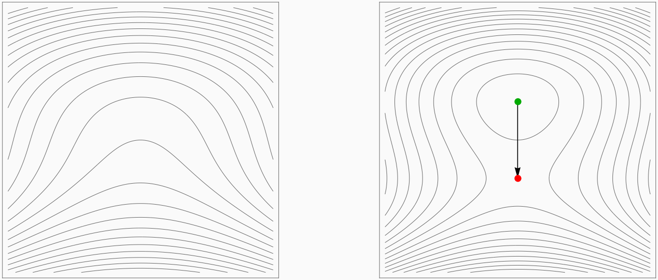


Figure 9: Critical points and gradient trajectories of birth-death bifurcation

In every picture, • is minimum, • is saddle, • is maximum.

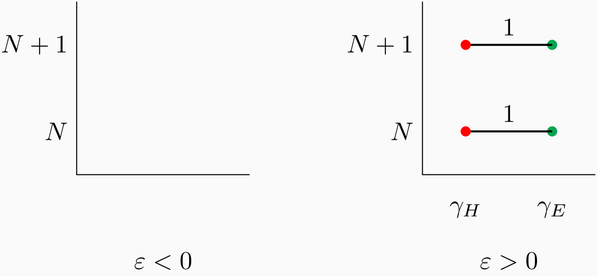


Figure 10: Chain complex mutation across the birth-death.

- Points at (n, m) corresponds to a generator of degree $n+m$.
- The line indicates the differential.
- We are using $\mathbb{Z}_2$-coefficients.

We have $HF_*^{\mathrm{loc}}(\gamma) = 0$, where γ is the degenerate orbit at $\varepsilon = 0$.

Period doubling $(k = 2)$

A generic double cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(u, v) = \alpha v^2 + \beta u^4 + \ldots + \varepsilon(u^2 + \ldots).$$

- $(\varepsilon < 0)$ One elliptic orbit γ_0^2 exists.
- $(\varepsilon > 0)$ γ_0^2 becomes hyperbolic, and an elliptic orbit ($\bullet$) γ_{new} is born.
- There is one cylinder between γ_0^2 and γ_{new}.

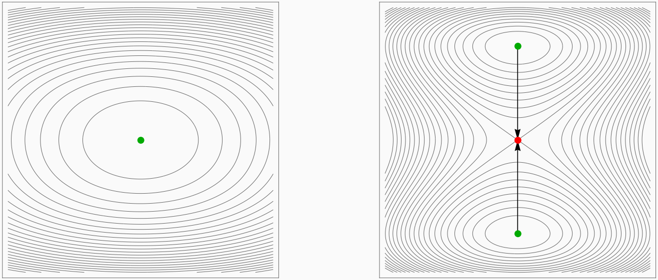


Figure 11: Period doubling bifurcation

Period doubling $(k = 2)$

- $\mu_{CZ}(\gamma_+^2; \gamma_-^2) = -1$ or 1. We assume -1 below.
- $\mu_{CZ}(\gamma_{new}; \gamma_-^2) = 0$.

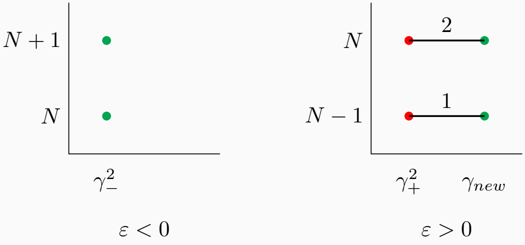


Figure 12: Chain complex mutation across the period doubling.

$\check{\gamma}_H$ and $\check{\gamma}^2$ cancel each other, while $\hat{\gamma}_H$ and $\hat{\gamma}^2$ do not.

$$HF_*^{\mathrm{loc}}(\gamma_0^2) \cong \begin{cases} \mathbb{Z}_2 & \text{if } * = 0, 1 \text{ relative to } \gamma_-^2, \\ 0 & \text{otherwise} \end{cases}$$

Phantom kiss $(k = 3, 4)$

A generic 3rd or 4th cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(\rho, \varphi) = \alpha\rho^k \cos k\varphi + \ldots + \varepsilon(\rho^2 + \ldots).$$

- $(\varepsilon < 0)$ One elliptic orbit γ^k and one hyperbolic orbit γ_H.
- $(\varepsilon = 0)$ They meet, and suddenly become unstable.
- $(\varepsilon > 0)$ The same configuration, but the index of γ^k changes.
- One cylinder exists between γ^k and γ_H.

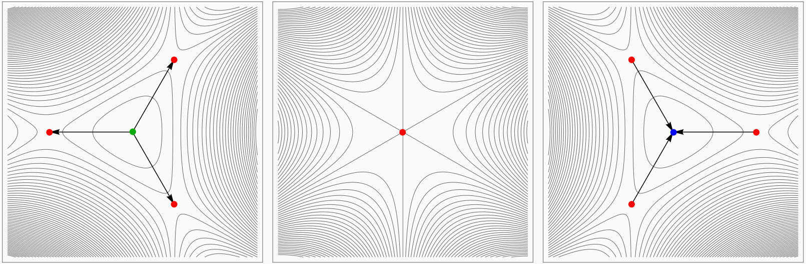


Figure 13: Phantom kiss bifurcation

Phantom kiss $(k = 3, 4)$

1. $\mu_{CZ}(\gamma_+^k; \gamma_-^k) = \pm 2$. Here, we assume $+2$.
2. $\mu_{CZ}(\gamma_{H,\pm}; \gamma_-^k) = \pm 1$, say -1, and it doesn't change.

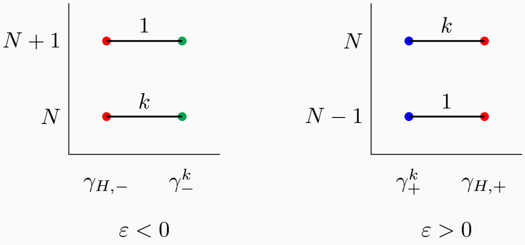


Figure 14: Chain complex mutations across the phantom kiss

$$HF_*^{\mathrm{loc}}(\gamma_0^3) = 0, \quad HF_*^{\mathrm{loc}}(\gamma_0^4) \cong \begin{cases} \mathbb{Z}_2 & \text{if } * = -1, 0 \text{ relative to } \gamma_-^4, \\ 0 & \text{otherwise} \end{cases}$$

Emission $(k \geq 4)$

Generic $k \geq 4$-th cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(\rho, \varphi) = \alpha\rho^4 + \beta\rho^k \cos k\varphi + \ldots + \varepsilon(\rho^2 + \ldots).$$

- $(\varepsilon < 0)$ One elliptic orbit γ_-^k exists.
- $(\varepsilon > 0)$ γ_-^k is still elliptic, and a new elliptic orbit γ_E and a new hyperbolic orbit γ_H are born.
- Two cylinders from γ_E to γ_H, and one cylinder from γ_H to γ_+^k.

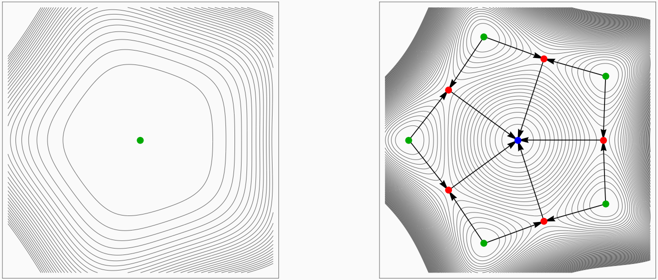


Figure 15: Emission bifurcation

Emission $(k \geq 4)$

1. $\mu_{CZ}(\gamma_+^k; \gamma_-^k) = \pm 2$, say -2.
2. $\mu_{CZ}(\gamma_H; \gamma_-^k) = -1$, $\mu_{CZ}(\gamma_E; \gamma_-^k) = 0$.

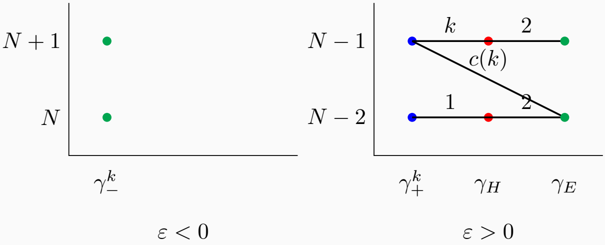


Figure 16: Chain complex mutations across the emission

$c(k)$ depends on the choice of Morse function, but always odd if k is even.

$$HF_*^{\text{loc}}(\gamma_0^k) \cong \begin{cases} \mathbb{Z}_2 & \text{if } * = 0, 1 \text{ relative to } \gamma_-^k, \\ 0 & \text{otherwise} \end{cases}$$

Emission $(k \geq 4)$

To find $c(k)$, we need to count the cascade of length 2 from $\check{\gamma}^k$ to $\hat{\gamma}_E$, which is equivalent to a count of Morse trajectories

$$\text{Image of } \check{\gamma}^k \text{ on } \gamma_H \;\rightarrow\; \text{Preimage of } \hat{\gamma}_E \text{ on } \gamma_H.$$

This falls into three cases, depending on the choice of the Morse function.

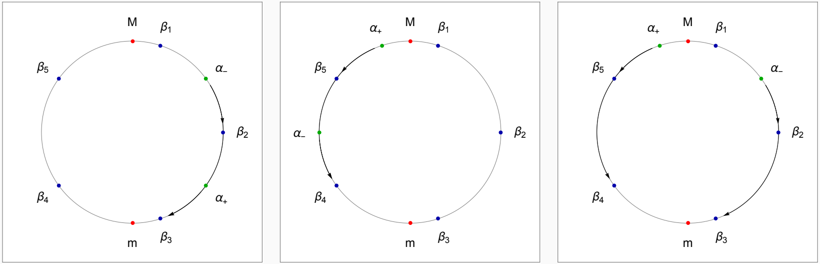


Figure 17: Three cases of $c(k)$.

Pitchfork ($\mathbb{Z}_2$-symmetric, $k = 1$)

For $\mathbb{Z}_2$-symmetric systems, we impose an extra $\mathbb{Z}_2$-symmetry on G_ε.

$\Rightarrow$ We analyze $\mathbb{Z}_{\mathrm{lcm}(2,k)}$-symmetric polynomials, so only odd-cover bifurcation produces new phenomena.

For $k = 1$, we have period doubling picture, where now two new critical points correspond to **two** new periodic orbits.

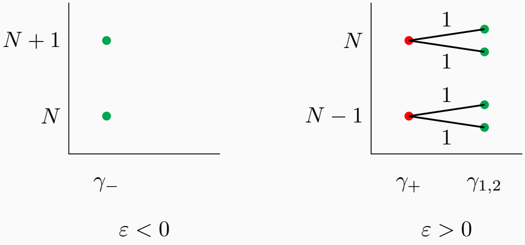


Figure 18: Chain complex mutations across the pitchfork

Double emission ($\mathbb{Z}_2$-symmetric, $k \geq 3$ odd)

For $k \geq 3$, we have $2k$-emission.

- There are two new hyperbolic and two new elliptic orbits.
- Cylinder configuration is changed. (One cylinder from each $\gamma_{E,i}$ to $\gamma_{H,j}$, and one cylinder from each $\gamma_{H,j}$ to γ^k.)

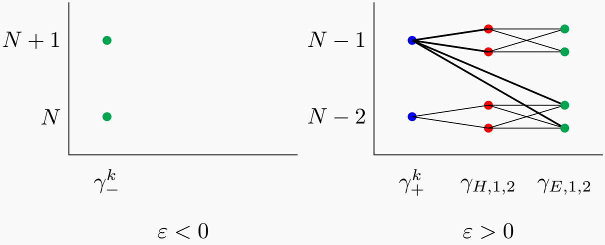


Figure 19: Chain complex mutations across the double emission

Further discussions

- Orientation and $\mathbb{Z}$-coefficients.
- More symmetries, multi-parameter, higher dimensional bifurcations.
- This gives a *generic* change of local Floer chain complex.
  $\Rightarrow$ Persistent module description and Floer-type Cerf theory?
- Bifurcation of symmetric orbits can be described as a Lagrangian Floer theory $\Rightarrow$ Similar work for Lagrangians?

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Figure 20: Thank you for your attention!

Questions, comments, and further discussion are welcome!